Smart Order Routers

The smart order router (SOR) is responsible for routing the child orders to the different exchanges, venues, and dark pools. The SOR will collect, monitor, and maintain trading activity data at the different venues and dark pools throughout the day using market/public data and in-house transactions. The SOR determines the likelihood of executing an order at each of the different venues based on frequency of trading and where the order would reside in the order book queue at that price. If the trading frequency and all else is equal across two venues, the SOR will route the limit order to the venue where it will sit highest in the queue. If one venue has 10,000 shares at the desired price and another venue has 5000 shares at the desired price, the SOR will route the shares to the venue with only 5000 shares since it is more likely to execute at that location quicker. The SOR determines on an expected value basis the likelihood of trading so it may route the order to a venue where it initially sits lower in the queue if that venue has higher trading activity or it may route the order to a venue that does not trade as frequently as others if it sits higher in the queue. For the most part this is an expected value calculation.

Revisiting our example above, where we are tasked with trading 1000 shares in the next 1 minute with the best mix of limit and market prices being: 200 @ \$29.95, 300 @ \$39.00, 300 @ \$30.05, and 200 @ \$30.10 (the offer), the SOR may decide to enter 200 shares in a dark pool at the midpoint of the bid-ask spread (\$30.05), 300 shares at the primary exchange at the best bid price of \$30.00. Furthermore, 200 shares may be entered into a non-traditional exchange @ \$29.95 where it sits first in the queue and perhaps even having to pay a rebate for posting (inverted pricing model).

Finally, the SOR may determine that to avoid potentially falling behind the schedule and having to possibly trade 200 shares at the market at the end of the 1 minute period where the price may move away (increase for the buy order) it would be best to trade 100 shares at the market immediately and 100 shares after perhaps 30−45 seconds, thus increasing the likelihood that they will not fall behind and have to trade in an aggressive and costly manner to catch up.

One reason why someone might decide to pay to post an order is to ensure that they will be the first 200 shares to trade at that price (if the market falls). Rational investors would route an order to a venue where they will receive a rebate to trade over a venue where they have to pay to trade. Here the savings achieved by transacting at the better price will more than make up for the rebate that has to be paid for posting liquidity.